Real Estate Management System report

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1. Introduction

The Real Estate Management System is a Java-based desktop application developed to help manage real estate operations in an organized and efficient way. The system was designed using Object-Oriented Programming (OOP) principles and connected to a MySQL database to store and manage records.

The application allows users to manage important real estate information such as properties, clients, agents, owners, and transactions through graphical user interfaces (GUI). The system also includes a login system for authentication, CRUD operations, and exporting transaction records into a text file.

The project aims to reduce manual work and provide an easy way to organize property management operations.

Problem Statement

Managing real estate information manually can lead to data loss, difficulty in tracking transactions, and poor organization of records. Real estate offices need a system that allows them to save, retrieve, update, and manage property information efficiently.

Objectives

The objectives of this project are:

- Build a real estate management application using Java.
- Apply Object-Oriented Programming concepts.
- Connect the application with a MySQL database.
- Create a user-friendly graphical interface.
- Allow users to add, display, update, and delete records.
- Implement a login system for user authentication.
- Export transaction data into a text file.
- Improve data organization and management.

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2. System Features

The Real Estate Management System provides several features to support real estate operations.

Login and Registration

The system allows users to create accounts and log in before accessing the dashboard. User authentication is implemented using the `UserService` class and database validation.

Property Management

The property management section allows users to:

- Add properties
- View properties
- Update property information
- Delete properties

The system supports three property types:

- Villa
- Apartment
- Land

Client Management

Users can:

- Add client information
- View stored clients
- Save client phone numbers and emails

Agent Management

Users can:

- Add agent information
- View agent records
- Manage supervision relationships between agents

Owner Management

Users can:

- Add owners
- View owner details
- Store owner contact information

Transaction Management

Users can:

- Save transactions
- View transaction history
- Store prices and related property information

- Export transaction data to a text file

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3. System Design

GUI Design

The system was developed using Java Swing to create graphical user interfaces.

The main interfaces of the system include:

Login Interface

The login interface allows users to enter username and password. If the information is correct, the dashboard will open.

Register Interface

The register screen allows new users to create accounts using:

- Name
- Email
- Password

The system validates user input before saving the account.

Dashboard Interface

The dashboard acts as the main navigation screen and contains buttons for:

- Properties
- Clients
- Agents
- Owners
- Transactions
- Exit

Property Interface

This interface allows users to:

- Add properties
- Show properties
- Update property information
- Delete properties

The property type can be selected from:

- Villa
- Apartment
- Land

Client Interface

The client interface stores:

- Client name
- Phone number
- Email

Agent Interface

The agent interface stores:

- Agent name
- First name
- Last name
- Supervisor ID

Owner Interface

The owner interface stores:

- Email
- First name
- Last name
- Phone number
- Owner name

Transaction Interface

The transaction interface stores:

- Date
- Final price
- Client ID
- Agent ID
- Property ID

It also includes an export button to save transaction records.

4. Class Hierarchy (Object-Oriented Programming)

The system applies Object-Oriented Programming concepts such as inheritance and abstraction.

The `Property` class is designed as an abstract superclass that stores common property information.

Property Class Attributes

- Property ID
- Location
- Status
- Property Type
- Owner ID
- Agent ID
- Price

Three subclasses inherit from the `Property` class:

Villa Class

The `Villa` class extends `Property` and represents villa properties.

Apartment Class

The `Apartment` class extends `Property` and represents apartment properties.

Land Class

The `Land` class extends `Property` and represents land properties.

Other classes included in the system are:

- User
- Client
- Agent
- Owner
- Transaction

Each class contains attributes and getter methods to retrieve data.

5. Database Design

The system is connected to a MySQL database called:

```
realestatemangmentsystem
```

The database stores information in several tables.

Users Table

Stores:

- User ID
- Name
- Email
- Password

Client Table

Stores:

- Client ID
- Client Name
- Phone Number
- Email

Agent Table

Stores:

- Agent ID
- Agent Name
- First Name
- Last Name
- Supervises

Owner Table

Stores:

- Owner ID
- Email
- First Name
- Last Name
- Phone Number
- Owner Name

Property Table

Stores:

- Property ID
- Location
- Status
- Property Type
- Owner ID
- Agent ID
- Price

Transaction Table

Stores:

- Transaction ID
- Date
- Final Price
- Client ID
- Agent ID
- Property ID

The tables are connected using IDs to maintain relationships between records.

6. Data Access Tools and Techniques

The project uses JDBC (Java Database Connectivity) to connect Java with MySQL.

The `DatabaseConnect` class is responsible for database connection using:

- Database URL
- Username
- Password

PreparedStatement is used to safely execute SQL commands.

The project implements CRUD operations:

Create

Insert data into:

- Users
- Clients

- Agents
- Owners
- Properties
- Transactions

Read

Retrieve stored records using SQL SELECT queries.

Update

Update property information using SQL UPDATE.

Delete

Delete properties using SQL DELETE.

The system also provides a login facility to control user access.

7. Error Handling

The system handles user errors using validation and exception handling.

GUI Validation

Input validation is implemented in the register screen.

Examples:

- Empty fields are not accepted.
- Email must contain `@`.
- Password must be at least 8 characters.

Exception Handling

The system uses `try-catch` blocks to prevent crashes and display error messages.

For example:

- Database connection errors
- Invalid input values
- SQL execution errors

Error messages are shown using `JOptionPane`.

8. Data Export

The system includes an export feature through the `ExportTransaction` class.

Transaction data is exported into:

```
transactions.txt
```

The exported file includes:

- Transaction ID
- Date
- Final Price

This feature helps users save and review transaction records outside the system.

9. Implementation

The application was implemented using:

- Java
- Java Swing
- MySQL
- JDBC

The project follows Object-Oriented Programming principles by organizing classes and services into separate files.

Service classes were implemented to communicate with the database, including:

- UserService
- ClientService
- AgentService
- OwnerService
- PropertyService
- TransactionService

The system structure improves code organization and maintainability.

10. Testing

The system was tested to ensure all functionalities work correctly.

Test Case 1: User Registration

Input: Valid name, email, and password
Expected Result: Account created successfully
Actual Result: Passed

Test Case 2: Login

Input: Correct username and password
Expected Result: Dashboard opens
Actual Result: Passed

Test Case 3: Add Client

Input: Client information
Expected Result: Client saved in database
Actual Result: Passed

Test Case 4: Add Property

Input: Property information
Expected Result: Property saved successfully
Actual Result: Passed

Test Case 5: Update Property

Input: Existing property ID with new data
Expected Result: Property updated successfully
Actual Result: Passed

Test Case 6: Delete Property

Input: Property ID
Expected Result: Property deleted successfully
Actual Result: Passed

Test Case 7: Export Transactions

Input: Click export button
Expected Result: File created successfully
Actual Result: Passed

11. Challenges Faced

Several challenges were faced during development, including:

- Connecting Java with MySQL database.
- Managing relationships between entities.
- Handling user input validation.
- Creating graphical interfaces using Java Swing.
- Implementing property inheritance using subclasses.

These problems were solved using debugging, exception handling, and proper class organization.

12. Conclusion

The Real Estate Management System successfully achieved its objectives by providing a complete desktop application for managing real estate information. The system applies Object-Oriented Programming concepts and database connectivity to organize records effectively.

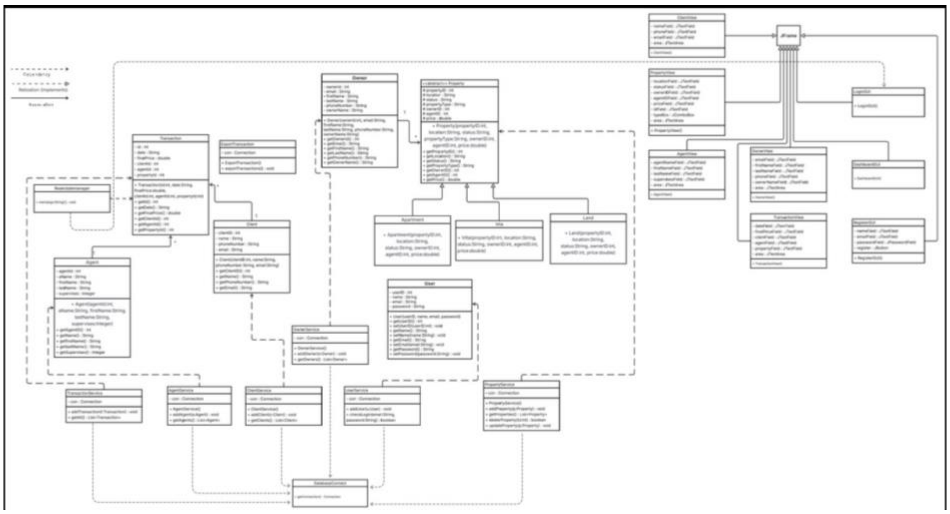
The application allows users to manage properties, clients, agents, owners, and transactions through easy-to-use graphical interfaces. It also includes login authentication, CRUD operations, validation, exception handling, and exporting transaction data.

Future Improvements

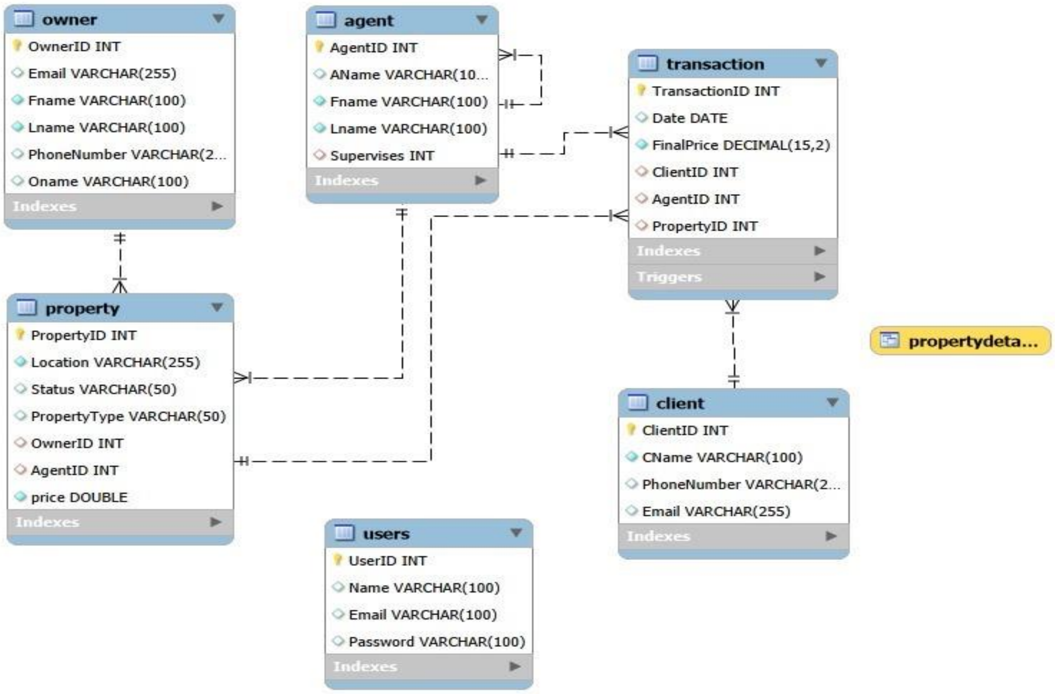
Future improvements may include:

- Adding search functionality
- Improving interface design
- Adding reports and statistics
- Enhancing database security
- Adding more property categories

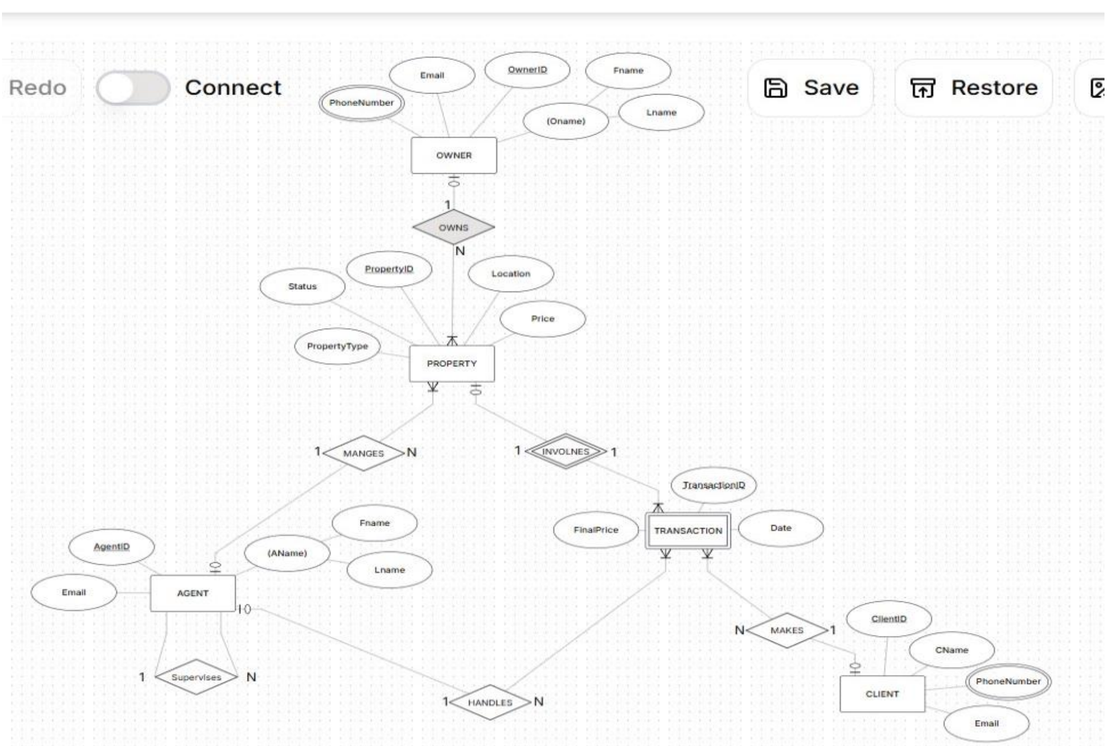
UML



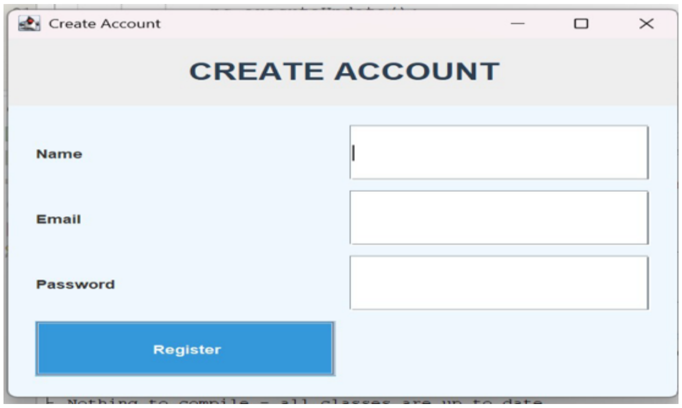
SQL QUERIE

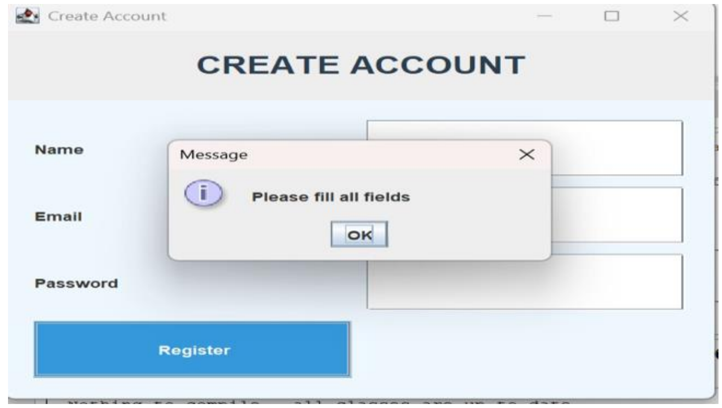


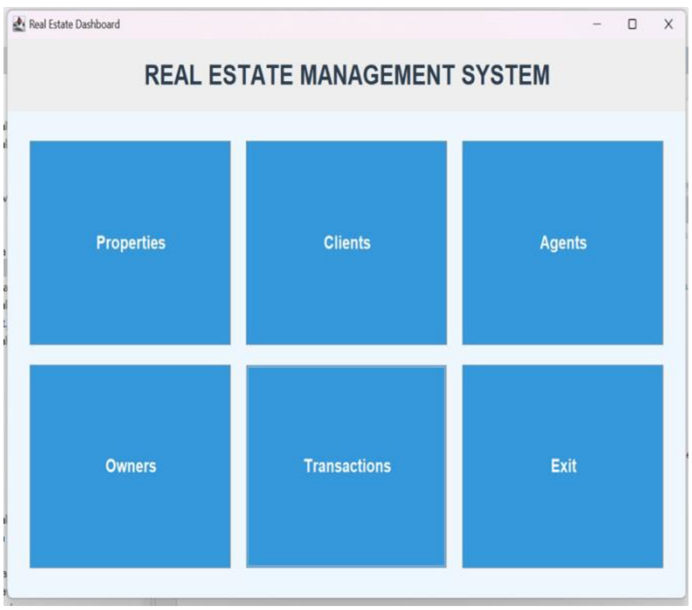
ERD

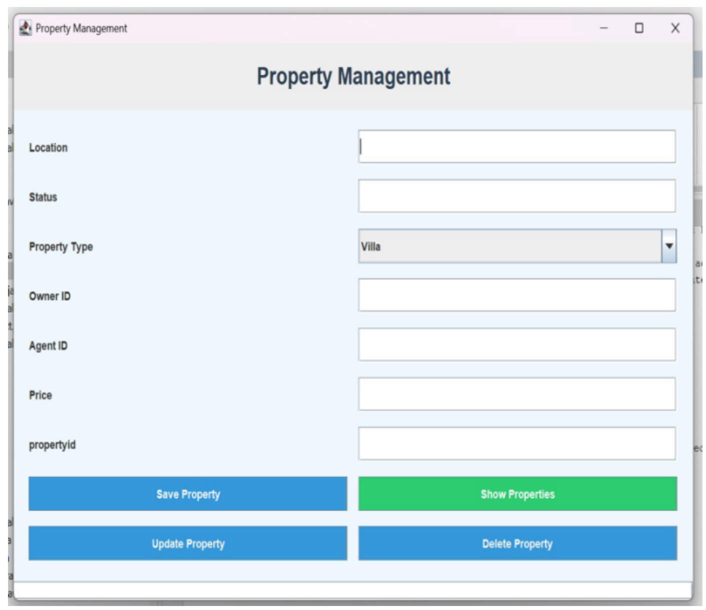


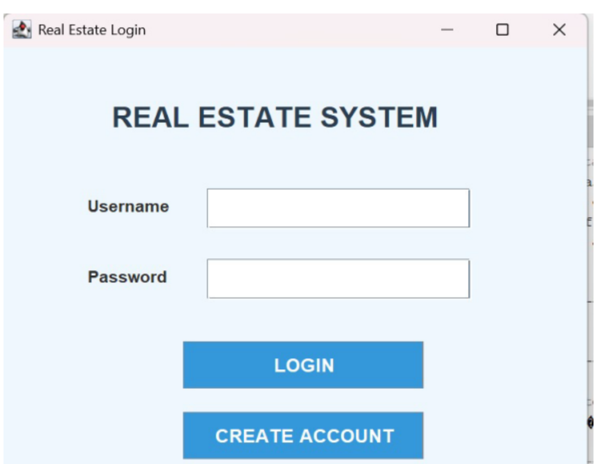
GUI Design

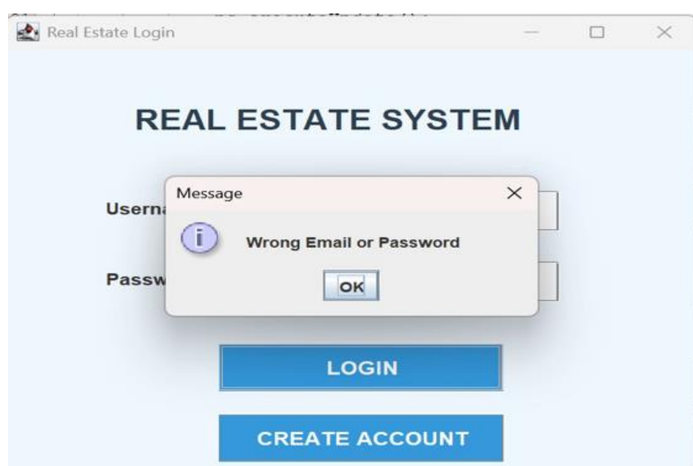

Create Account
CREATE ACCOUNT
Name
Email
Password
Register

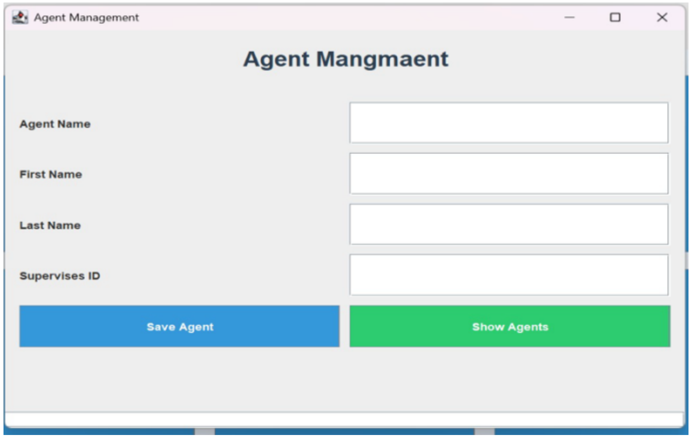

Create Account
CREATE ACCOUNT
Name
Email
Password
Message
Please fill all fields
OK
Register

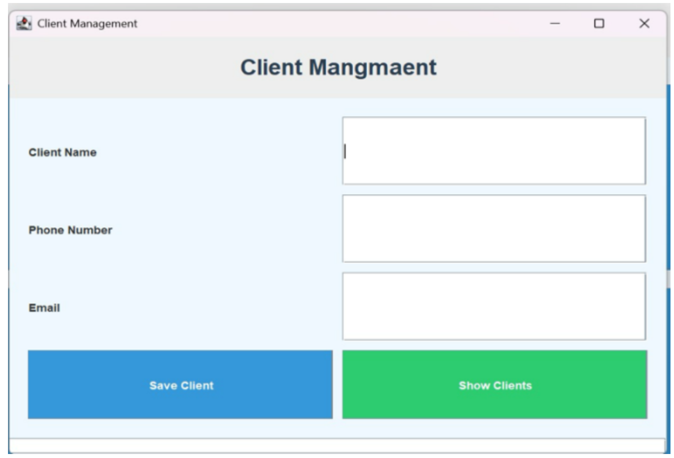

Real Estate Dashboard
REAL ESTATE MANAGEMENT SYSTEM
Properties
Clients
Agents
Owners
Transactions
Exit


Property Management
Property Management
Location
Status
Property Type
Villa
Owner ID
Agent ID
Price
propertyid
Save Property
Show Properties
Update Property
Delete Property

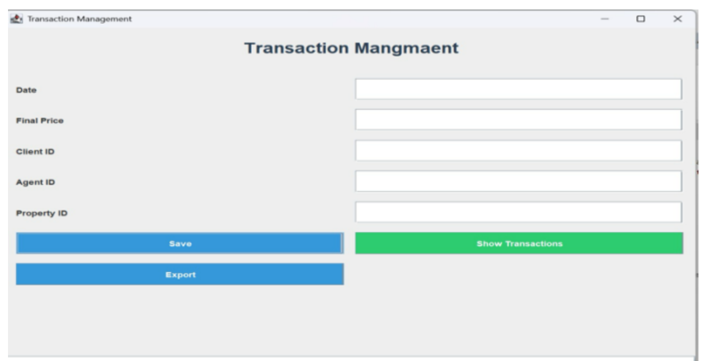

Real Estate Login
REAL ESTATE SYSTEM
Username
Password
LOGIN
CREATE ACCOUNT


Real Estate Login
REAL ESTATE SYSTEM
Message
Wrong Email or Password
OK
LOGIN
CREATE ACCOUNT


Agent Management
Agent Mangmaent
Agent Name
First Name
Last Name
Supervises ID
Save Agent
Show Agents


Client Management
Client Mangmaent
Client Name
Phone Number
Email
Save Client
Show Clients


Owner Management
Owner Mangmaent
Email
First Name
Last Name
Phone Number
Owner Name
Save Owner
Show Owners


Transaction Management
Transaction Mangmaent
Date
Final Price
Client ID
Agent ID
Property ID
Save
Show Transactions
Export